

Exam Date & Time: 20-Nov-2024 (02:30 PM - 05:30 PM)



## MANIPAL ACADEMY OF HIGHER EDUCATION

MIT BLR - BTech VII Semester - End Semester Examination - Nov / Dec -2024

### MACHINE LEARNING FOR DATA ANALYTICS [IT-4056]

Marks: 50

Duration: 180 mins.

#### PART-A

Answer all the questions.

1) Refer Table 1A to answer the following

A)

| Customer_ID | Age | Purchase_Amount | Purchase_Date | Product_Category |
|-------------|-----|-----------------|---------------|------------------|
| C001        | 25  | 120.5           | 15-01-2024    | Electronics      |
| C002        | 30  | 55              | 17-01-2024    | Clothing         |
| C003        | 22  | 80              | 20-01-2024    | Home Goods       |
| C004        | 28  | 200             | 25-01-2024    | Electronics      |
| C005        | 35  | -10             | 01-02-2024    | Clothing         |
| C006        | NaN | 150             | 05-02-2024    | Electronics      |
| C007        | 27  | 70              | 10-02-2024    | Home Goods       |
| C008        | 33  | 90              | 15-02-2024    | Clothing         |
| C009        | 29  | 250             | 01-03-2024    | Electronics      |
| C010        | 31  | 60              | 05-03-2024    | Home Goods       |

(5)

Table 1A

a. Identify and explain two data quality issues present in the dataset. How might these issues affect any analysis conducted on this data?

b. Draft a data quality report based on the dataset. Address the presence of any issues related to the dataset's integrity and consistency. Propose appropriate measures for addressing these issues.

B)

Demonstrate what can go wrong when an inappropriate inductive bias is used?

(3)

C)

(2)

| ID | Target | Prediction | ID | Target | Prediction | ID | Target | Prediction |
|----|--------|------------|----|--------|------------|----|--------|------------|
| 1  | false  | false      | 8  | true   | true       | 15 | false  | false      |
| 2  | false  | false      | 9  | false  | false      | 16 | false  | false      |
| 3  | false  | false      | 10 | false  | false      | 17 | true   | false      |
| 4  | false  | false      | 11 | false  | false      | 18 | true   | true       |
| 5  | true   | true       | 12 | true   | true       | 19 | true   | true       |
| 6  | false  | false      | 13 | false  | false      | 20 | true   | true       |
| 7  | true   | true       | 14 | true   | true       |    |        |            |

**Table 1C**

For data set in Table 1C,  
Calculate precision, recall, and F1 measure

- 2) Table 2a shows the policy type held by customers at a life insurance company.

A)

| ID | POLICY   | ID | POLICY   | ID | POLICY   |
|----|----------|----|----------|----|----------|
| 1  | Silver   | 8  | Silver   | 15 | Platinum |
| 2  | Platinum | 9  | Platinum | 16 | Silver   |
| 3  | Gold     | 10 | Platinum | 17 | Platinum |
| 4  | Gold     | 11 | Silver   | 18 | Platinum |
| 5  | Silver   | 12 | Gold     | 19 | Gold     |
| 6  | Silver   | 13 | Platinum | 20 | Silver   |
| 7  | Bronze   | 14 | Silver   |    |          |

(5)

**Table2a**

a. Based on this data calculate the following summary statistics for the POLICY feature:

- Mode and 2nd mode
  - Mode % and 2nd mode %
- b. Draw a bar plot for the POLICY feature.

B)

(3)

Table 2b gives details of symptoms that patients presented and whether they were suffering from meningitis.

| ID | HEADACHE | FEVER | VOMITING | MENINGITIS |
|----|----------|-------|----------|------------|
| 1  | true     | true  | false    | false      |
| 2  | false    | true  | false    | false      |
| 3  | true     | false | true     | false      |
| 4  | true     | false | true     | false      |
| 5  | false    | true  | false    | true       |
| 6  | true     | false | true     | false      |
| 7  | true     | false | true     | false      |
| 8  | true     | false | true     | true       |
| 9  | false    | true  | false    | false      |
| 10 | true     | false | true     | true       |

**Table 2b**

Using this dataset calculate the following probabilities:

- $P(\text{VOMITING} = \text{true})$
- $P(\text{HEADACHE} = \text{false})$
- $P(\text{HEADACHE} = \text{true}, \text{VOMITING} = \text{false})$
- $P(\text{VOMITING} = \text{false} \mid \text{HEADACHE} = \text{true})$
- $P(\text{MENINGITIS} \mid \text{FEVER} = \text{true}, \text{VOMITING} = \text{false})$

C) Outline the constraints based on which, the algorithm defines three situations where the recursion stops, and a leaf node is constructed in ID3 algorithm. (2)

3) (5)

A)

A convicted criminal who reoffends after release is known as a recidivist. Table 3a lists a dataset that describes prisoners released on parole, and whether they reoffended within two years of release

Fig 3A

| ID | GOOD<br>BEHAVIOR | AGE <<br>30 | DRUG<br>DEPENDENT | RECIDIVIST |
|----|------------------|-------------|-------------------|------------|
| 1  | FALSE            | TRUE        | FALSE             | TRUE       |
| 2  | FALSE            | FALSE       | FALSE             | FALSE      |
| 3  | FALSE            | TRUE        | FALSE             | TRUE       |
| 4  | TRUE             | FALSE       | FALSE             | FALSE      |
| 5  | TRUE             | FALSE       | TRUE              | TRUE       |
| 6  | TRUE             | FALSE       | FALSE             | FALSE      |

Table 3a

Apply the ID3 algorithm to the dataset to determine the resulting root node split using

- Entropy-based information gain.
- Gini index-based information gain.

Provide all necessary computations for each step of the process to support your findings.

B) Describe different Measuring Similarities Using Distance Metrics. (3)

C) In some data set the test dataset is quite imbalanced, containing 90 instances with the non-churn level and just 10 instances with the churn level. This means that the performance of the model on the non-churn level overwhelms the performance on the churn level in the accuracy calculation and illustrates how classification accuracy can be a misleading measure of model performance. (2)

4) (5)

A)

The use of the kernel trick is key in writing efficient implementations of the support vector machine approach to predictive modelling. The kernel trick is based on the fact that the result of a kernel function applied to a support vector and a query instance is equivalent to the result of calculating the dot product between the support vector and the query instance after a specific set of basis functions have been applied to both—in other words,  $\text{kernel}(\mathbf{d}, \mathbf{q}) = \phi(\mathbf{d}) \cdot \phi(\mathbf{q})$ .

Using the support vector  $\langle \mathbf{d}[1], \mathbf{d}[2] \rangle$  and the query instance  $\langle \mathbf{q}[1], \mathbf{q}[2] \rangle$  as examples, show that applying a polynomial kernel with  $p=2$ ,  $\text{kernel}(\mathbf{d}, \mathbf{q}) = (\mathbf{d} \cdot \mathbf{q} + 1)^2$ , is equivalent to calculating the dot product of the support vector and query instance after applying the following set of basis functions:

$$\begin{aligned} \phi_0(\langle \mathbf{d}[1], \mathbf{d}[2] \rangle) &= \mathbf{d}[1]^2 & \phi_1(\langle \mathbf{d}[1], \mathbf{d}[2] \rangle) &= \mathbf{d}[2]^2 \\ \phi_2(\langle \mathbf{d}[1], \mathbf{d}[2] \rangle) &= \sqrt{2} \times \mathbf{d}[1] \times \mathbf{d}[2] & \phi_3(\langle \mathbf{d}[1], \mathbf{d}[2] \rangle) &= \sqrt{2} \times \mathbf{d}[1] \\ \phi_4(\langle \mathbf{d}[1], \mathbf{d}[2] \rangle) &= \sqrt{2} \times \mathbf{d}[2] & \phi_5(\langle \mathbf{d}[1], \mathbf{d}[2] \rangle) &= 1 \end{aligned}$$

- B) Designing a fraud detection system for a financial institution requires addressing the challenge of detecting rare fraudulent transactions among a large number of normal ones. Considering the class imbalance, determine whether a generative or discriminative model would be more suitable. Provide justification based on how the model handles class imbalance and the need for interpretability. (3)
- C) Illustrate the a high classification accuracy be misleading as an evaluation metric? Provide an example where other metrics might provide more insight. (2)
- 5) (5)
- A)

A support vector machine model has been trained to differentiate between dosages of two drugs that either cause a dangerous interaction or interact safely. The model utilizes two continuous features, DOSE1 and DOSE2, with target levels classified as dangerous (positive level, +1) and safe (negative level, -1). The support vectors from the trained model are presented in Table 5B.

| DOSE1   | DOSE2   | CLASS |
|---------|---------|-------|
| 0.2351  | 0.4016  | 1     |
| -0.1764 | -0.1916 | 1     |
| 0.3057  | -0.9394 | -1    |
| 0.559   | 0.6353  | -1    |
| -0.66   | -0.1175 | -1    |

Table 5B

In the trained model the value of  $w_0$  is 0.3074, and the values of the  $\alpha$  parameters are (7.1655, 6.9060, 2.0033, 6.1144, 5.9538).

Given a query instance with DOSE1 = 0.90 and DOSE2 = -0.90, calculate the output of the support vector machine model using the version that incorporates basis functions (refer to Equation 5B)

Equation 5B:

$$M_{a,\phi,w_0}(q) = \sum_{i=1}^s (t_i \times a[i] \times (\phi(d_i) \cdot \phi(q)) + w_0)$$

- B) For each model identify whether they are parametric or non-parametric and whether they are generative or discriminative:
- Logistic Regression
  - Bayesian Network
  - SVM
- (3)
- C) An online movie streaming company has a business problem of growing customer churn-subscription customers cancelling their subscriptions to join a competitor. Create a list of ways in which predictive data analytics could be used to help address this business problem. For each proposed approach, describe the predictive model that will be built, how the model will be used by the business, and demonstrate how using the model will help address the original business problem.
- (2)

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